

JACOB W. MURRI

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SUMMARY

Applied mathematician (Ph.D. expected 2027, UCLA) developing data-driven methods for nonlinear partial differential equations (PDEs) in computational fluid dynamics (CFD): data assimilation, model calibration, reduced-order and surrogate modeling, and scientific machine learning (SciML), in parallel codes for high-performance computing (HPC). Research experience at the National Security Agency, Nevada National Security Sites, and Institute for Defense Analyses.

EDUCATION

Ph.D., Mathematics

expected 2027

University of California, Los Angeles

Dissertation: *Learning States and Interactions in Nonlinear Partial Differential Equations with Data* (working title). Advisor: Hayden Schaeffer. Qualifying examinations in Applied Differential Equations, Numerical Analysis, and Optimization and Numerical Linear Algebra.

M.S., Mathematics

2022

Brigham Young University

Thesis: *Data Assimilation and Parameter Recovery for Rayleigh–Bénard Convection*. Advisor: Jared P. Whitehead.

B.S., Mathematics, *summa cum laude*

2021

Brigham Young University

AWARDS AND HONORS

Liggett Teaching Award, UCLA (2026) • Honorable Mention, NSF Graduate Research Fellowship (2022) • Dean's Fellowship, BYU (2021–2022) • Orson Pratt Prize in Mathematics, BYU (2021)

RESEARCH AND PROFESSIONAL EXPERIENCE

Graduate Researcher

September 2022–present

Department of Mathematics, University of California, Los Angeles

- Reconstructed unobserved velocity fields of buoyancy-driven flows from sparse temperature sensors alone using continuous data assimilation (state estimation), in an NNSS-funded collaboration with Nevada National Security Sites scientists.
- Discovered that this reconstruction succeeds only within a window of the relaxation parameter, contrary to standard theory, and that the window closes at high Rayleigh number; explained the failure mechanism with a singular perturbation analysis and proved a local synchronization theorem.
- Developed an MPI-parallel FEniCSx/DOLFINx Boussinesq solver on unstructured meshes (projection scheme, PETSc Krylov solvers, algebraic multigrid) and ran parameter studies spanning 25 flow regimes and several orders of magnitude in the relaxation parameter on Slurm-managed HPC clusters.
- Calibrated a nonlocal PDE model of bacterial colony growth to experimental image sequences with Montana State University experimentalists, solving a PDE-constrained inverse problem to recover interaction kernels that cannot be derived from first principles.

Adjunct Research Staff Member

June–August 2026

Institute for Defense Analyses, La Jolla, California

- Developed loss-aware decoding algorithms for qLDPC codes on neutral atom quantum computers exploiting correlated loss from two-qubit gates
- Developed procedures and implemented code for simulating atom loss events in quantum syndrome extraction circuits using Stim
- Used an agentic coding workflow with Claude Code to speed up development of software for testing research ideas
- Communicated results to experts and non-experts in internal presentations, informal reports, and refereed technical papers

Associate in Science

May–August 2024

Special Technologies Laboratory, Nevada National Security Sites, Santa Barbara, CA

- Contributed to a Site-Directed Research and Development program in computational fluid dynamics (CFD).
- Developed a hybrid reduced-order model (ROM) / full-order model for incompressible flow around obstacles, coupled through a discontinuous Galerkin domain decomposition with $H(\text{div})$ -conforming finite elements and advanced by a least-squares Petrov–Galerkin (LSPG) scheme.
- Trained proper orthogonal decomposition (POD) reduced bases on small subdomains that transfer to obstacle configurations unseen in training, reproducing full-order flow at reduced time and memory cost.

Applied Research Mathematician

May–August 2021

National Security Agency, Fort Meade, MD

- Contributed to analysis of cybersecurity of algorithms for the NIST post-quantum cryptography standardization effort.
- Compiled findings into an internally refereed technical paper.
- Presented results at Institute for Defense Analyses centers in Princeton, NJ and Bowie, MD.
- Obtained a TS/SI security clearance.

Undergraduate Research Assistant

2019–2021

Department of Mathematics, Brigham Young University

- Proved a stability criterion for stochastically switched and time-delayed dynamical systems, and used Markov chain Monte Carlo sampling to quantify gerrymandering in Utah redistricting plans; both projects led to journal publications.

TECHNICAL SKILLS

Programming and HPC: Python (NumPy, SciPy, Matplotlib, PyTorch, Dedalus, FEniCSx/DOLFINx, petsc4py, mpi4py), MPI, PETSc, Slurm, bash, git, L^AT_EX.

Numerical methods: finite element method (FEM), discontinuous Galerkin (DG), spectral, and pseudospectral methods for nonlinear PDEs; projection methods; algebraic multigrid; Anderson acceleration; verification and validation.

Data-driven modeling: data assimilation; state and parameter estimation; model calibration; inverse problems; uncertainty quantification; derivative-free and gradient-based PDE-constrained optimization; reduced-order models (ROMs) with proper orthogonal decomposition (POD); surrogate modeling; scientific machine learning.

Languages: English (native), Indonesian (fluent).

PUBLICATIONS

- **J. W. Murri**, S. Breckling, C. Watkins, and H. Schaeffer. Temperature-only data assimilation in mixed convection: nonmonotone dependence on the relaxation parameter. In preparation.
- **J. W. Murri**, C. Dudiak, J. N. Wilking, S. G. McCalla, and H. Schaeffer. Learning interaction kernels for nonlocal bacterial growth models. In preparation.
- **J. W. Murri**, S. Breckling, and C. Watkins. Submerged reduced-order models for incompressible flow around obstacles. In preparation.
- V. R. Martinez, **J. W. Murri**, and J. P. Whitehead. Relaxation-based schemes for on-the-fly parameter estimation in dissipative dynamical systems. *Inverse Problems*, 41(5):055001, 2025.
- A. King, **J. W. Murri**, J. Callahan, A. Russell, and T. J. Jarvis. Mathematical analysis of redistricting in Utah. *Statistics and Public Policy*, 9(1):136–148, 2022.
- C. Carter, **J. W. Murri**, D. Reber, and B. Webb. A simple stability criterion for dynamical systems with stochastic switching and/or stochastic time-delays. *Nonlinearity*, 35(12):6042, 2022.

SELECTED PRESENTATIONS

- Recovering mixed convection flows using only temperature sensor measurements. Department of Mathematical Sciences Colloquium, Montana State University, April 2026 (invited).
- Recovering mixed convection flows using only temperature sensor measurements. Southern California Applied Mathematics Symposium, April 2026.
- Exploring continuous data assimilation algorithms for reconstructing flow from temperature measurements. APS Division of Fluid Dynamics Meeting, November 2025.
- Learning interaction kernels for nonlocal bacterial growth models (poster). IPAM Workshop on Sampling, Inference, and Data-Driven Physical Modeling in Scientific Machine Learning, July 2025.
- Submerged reduced-order models for incompressible flow around obstacles. APS Division of Fluid Dynamics Meeting, November 2024.

TEACHING AND MENTORING

- **Instructor of record**, Mathematical Methods for Data (UCLA, 2026) and Essentials of Calculus (BYU, 2022); **teaching assistant** for 16 UCLA course offerings (2022–2026), including applied numerical methods, machine learning, analysis, and applied ODEs.
- **Research mentor**, UCLA Computational and Applied Mathematics REU (2023), for six undergraduates modeling bacterial colony growth; **graduate mentor**, UCLA Directed Reading Program (2025).